

MLDM Tutorial 7 solution

- 1) Fill out the confusion matrix for this data and calculate the accuracy, precision, recall and F1 Score

Patient Number	Does the patient have the disease?	Does the model predict they have it?
1	Yes	Yes
2	Yes	No
3	No	No
4	No	Yes
5	Yes	No
6	No	No

True Positives (TP): 1 (Patient 1)

False Positives (FP): 1 (Patient 4)

True Negatives (TN): 2 (Patients 3 and 6)

False Negatives (FN): 2 (Patients 2 and 5)

	Predicted: Yes	Predicted: No
Actual: Yes	TP = 1	FN = 2
Actual: No	FP = 1	TN = 2

Precision (also known as Positive Predictive Value):

$$Precision = \frac{TruePositives}{TruePositives+FalsePositives}$$

Recall (also known as Sensitivity or True Positive Rate):

$$Recall = \frac{TruePositives}{TruePositives+FalseNegatives}$$

F1 Score (the harmonic mean of precision and recall):

$$F1Score = \frac{2 \cdot Precision \cdot Recall}{Precision + Recall}$$

1. **Precision:**

$$Precision = \frac{1}{1+1} = \frac{1}{2} = 0.5$$

2. **Recall:**

$$Recall = \frac{1}{1+2} = \frac{1}{3} \approx 0.3333$$

3. **F1 Score:**

$$F1\ Score = \frac{2 \cdot 0.5 \cdot 0.3333}{0.5 + 0.3333} \approx \frac{0.3333}{0.8333} \approx 0.4$$

2) What is the difference of filter, wrapper and embedded methods? What are the advantage and disadvantage of each approach?

Filter methods involve selecting features based on their statistical properties, such as correlation with the target variable, information gain, or mutual information. These methods do not involve training a model but rather analyze feature characteristics independently.

Advantages:

Computationally efficient and faster, making them suitable for high-dimensional datasets.

Feature selection is independent of the machine learning algorithm.

Can provide insights into feature relevance.

Disadvantages:

May not capture complex feature interactions.

Ignores the impact of feature selection on model performance.

Wrapper Methods: select features by evaluating the performance of a machine learning model trained on different subsets of features. They use the model's predictive ability to determine feature relevance.

Advantages:

Considers feature interactions and their impact on model performance.

Can lead to better feature subsets for specific models.

Disadvantages:

Computationally intensive, as it requires training and evaluating the model multiple times.

Prone to overfitting when not using proper cross-validation techniques.

May not be suitable for high-dimensional datasets due to computational demands.

Embedded methods incorporate feature selection within the model-building process. Features are selected or weighted as part of model training.

Advantages:

Efficient feature selection as part of model training.

Tends to perform well with datasets having many features.

Considers feature interactions and can adapt to various machine learning algorithms.

Disadvantages:

Depending on the size of the dataset and the complexity of the embedded feature selection method, the process can be computationally demanding. This can make embedded methods impractical for large datasets or real-time applications.

Just like wrapper methods, embedded methods are also susceptible to overfitting if not used with appropriate cross-validation techniques

They may not explore the full feature space as thoroughly as wrapper methods, which consider various feature combinations.

In summary:

Filter methods are computationally efficient but might miss feature interactions.

Wrapper methods involve model performance but are computationally intensive and may lead to overfitting.

Embedded methods provide an intermediate approach that can be efficient and effective, especially with high-dimensional data.

The choice of method depends on the dataset, the machine learning algorithm, and the specific problem you are working on. It's often a good practice to try multiple methods and compare their performance to select the most appropriate feature selection approach.

- 3) Which method of feature selection employs statistical measures to identify the most relevant features?
- a) Wrapper method
 - b) Embedded method
 - c) Filter method
 - d) Hybrid method

Answer: c

- 4) What does the "wrapper method" in feature selection rely on?
- a) Statistical measures
 - b) Feature importance ranking
 - c) Classifier's performance
 - d) Correlation coefficients

Answer: c

- 5) Which of the following best defines feature selection in machine learning?
- a) The process of creating new features from existing ones.
 - b) The process of choosing the most relevant features from a dataset.
 - c) The process of training a model with all available features.
 - d) The process of transforming categorical features into numerical ones.

Answer: b

- 6) Which technique involves discarding instances from the majority class to balance class distribution?
- a) Oversampling
 - b) Undersampling

- c) SMOTE
- d) Synthetic sampling

Answer: b

- 7) Which method involves creating artificial samples for the minority class using characteristics of existing samples?
- a) Oversampling
 - b) Undersampling
 - c) Random sampling
 - d) SMOTE

Answer: d

- 8) What does stratified cross-validation aim to achieve?
- a) It aims to perform feature selection.
 - b) It aims to create a balanced class distribution in each fold.
 - c) It aims to reduce overfitting.
 - d) It aims to increase computational efficiency.

Answer: b

- 9) Which cross-validation method is computationally expensive and is prone to overfitting?
- a) Leave-One-Out Cross-Validation (LOOCV)
 - b) K-Fold Cross-Validation
 - c) Stratified Cross-Validation
 - d) Hold-Out Cross-Validation

Answer: a

- 10) What is the primary goal of hyperparameter tuning?
- a) To maximize the number of features used in the model.
 - b) To minimize the time required for training the model.
 - c) To ensure that the model always converges to the optimal solution.
 - d) To optimize the model's performance by finding the best hyperparameters.

Answer: d